

Radix 4

plain binary — one digit per pair of bits of B

Step 1 — the number

B is an n -bit two's complement value (n even). The top bit counts negative:

$$B = -2^{n-1} B^{n-1} + \sum_{j=0}^{n-2} 2^j B^j$$

Step 2 — group the bits in pairs

Each pair of bits is one digit. The top pair holds the negative bit:

$$B = \sum_{j=0}^{n/2-1} 4^j d_j \quad \begin{array}{l} d_j = 2 B^{2j+1} + B^{2j} \in \{0, 1, 2, 3\} \\ d_{n/2-1} = -2 B^{n-1} + B^{n-2} \in \{-2, -1, 0, 1\} \end{array} \quad j < n/2 - 1$$

Step 3 — the product

$$AB = \sum_{j=0}^{n/2-1} 4^j d_j A$$

$d_j A$ takes one of four values in each position:

B^{2j+1}	B^{2j}	lower digits		top digit	
		d_j	$d_j A$	$d_{n/2-1}$	$d_{n/2-1} A$
0	0	0	0	0	0
0	1	1	A	1	A
1	0	2	$2A$	-2	$-2A$
1	1	3	$3A$	-1	$-A$

$n/2$ digits, $n/2$ partial products. The digit 3 is the problem: $3A = 2A + A$ needs an adder per partial product. Booth and NR4SD are two ways of removing it.